

ECE 2210: CI: Introduction to Programming in Python

HW2 – Arithmetic Operators and Functions

Complete the **four** functions given in the Python scaffold `hw2.py`. The headers of all four functions are given. The requirements of completing these functions are given in the triple-quoted comments above each function. The following information is needed to complete the coding. Note that you are **NOT** allowed to use scientific notation in your code.

- a) (40 pts) Function I & II: `cal_secs_ball_falling1()` and `cal_secs_ball_falling2()`

The standard kinematic formula is $s = \frac{1}{2}gt^2$, where $g = 9.81\text{ms}^{-2}$ is the acceleration due to gravity.

- b) (30 pts) Function III: `cal_altitude()`

The altitude h above the earth's surface that the satellite must have is given by

$$h = \left(\frac{GMT^2}{4\pi^2} \right)^{1/3} - R$$

where $G = 6.67 \times 10^{-11} \text{ m}^3 \text{ kg}^{-1} \text{ s}^{-2}$ is Newton's gravitational constant, $M = 5.97 \times 10^{24} \text{ kg}$ is the mass of the Earth, and $R = 6371 \text{ km}$ is its radius. The unit of h is m (**not** km).

- c) (30 pts) Function IV: `draw_grid()`

The following graph shows the grid printed by function “`draw_grid`”. You are **required** to use string multiplication and string concatenation for this one. You will receive 0 for this problem without proper use of the related string operations.

```
+ - - - - + - - - - +
|           |           |
|           |           |
|           |           |
+ - - - - + - - - - +
|           |           |
|           |           |
|           |           |
+ - - - - + - - - - +
```

Hint: to print more than one value on a line, you can print a comma-separated sequence of values:

```
print('+', '-')
```

By default, `print` advances to the next line, but you can override that behavior and put a space at the end, like this:

```
print('+', end = ' ')
print('-')
```

The output of these statements is ‘+ -’. A `print` statement with no argument ends the current line and goes to the next line.

Code testing:

The correct output is given in file “`output`”. You can use any text editor (e.g., Sublime) to open it to see what the correct output looks like. To test your code, run it in the Linux terminal

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using the following command, which generates a text file called “**myoutput**” that contains the output of your program.

```
$ python3 hw2.py > myoutput
```

Then, run the following command in Linux terminal to compare the output of your program and the given output.

```
$ diff output myoutput
```

If there is no output of running **diff**, it means the **output** and **myoutput** are identical. Note that TA will use the same command to test your program.

Strict Output Matching: Your program’s output must match the provided output file **EXACTLY** in both values and formatting (including spelling, punctuation, capitalization, spaces, and line structure). Any deviation in text formatting (e.g., extra prompt text, misspelled words, incorrect spacing in the grid, or unexpected print statements) will result in a **score of 0** for that function’s correctness test.

Submission: For the submission of this assignment, you need to upload the completed **hw2.py** to Canvas.

Grading: The output(s) of each function is worth 25 points.

Warning: The TA will compare your code with and your classmates’ codes using MOSS (<https://theory.stanford.edu/~aiken/moss/>). A similarity of 40% or higher is considered as plagiarism. The best way to avoid a high similarity is to write the code all by yourself. Discussion is always encouraged but never look at others’ code.